Ricardo Miguel da Silva Ribeiro

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Education

Instituto Superior Técnico *Sep 2026 - Jun 2028*

Master's Degree in Computer Science and Engineering

- Specializing in Artificial Intelligence and Distributed Systems.

Instituto Superior Técnico *Sep 2023 - Jun 2026*

Bachelor's Degree in Engineering Physics

- Final Grade: 16.02/20

Research Experience

Research Engineer for Noisr *Jan 2024 - Jun 2026*

- Led the development of the learning algorithm for an electronics-based Physical Neural Network, while also contributing to an optics-based approach.
- Designed and prototyped crossbar and omega network architectures for analogue computing circuits.
- Received the LEFT/MEFT Student Innovation Prize at the IST-Physics Frontier Symposium.
- Contributed to the development of a C++ library, implementing statistical tests from the ENT test suite to verify the reliability of physical random number generators.
- Presented our research in outreach events organised by the Physics Department.

Research Intern for GoLP *Sep 2025 - Feb 2026*

- Developed a generalized quantum simulation framework for coupled harmonic oscillators, encoding classical dynamics into quantum Hamiltonians and implementing Suzuki-Trotter time evolution.
- Benchmarked quantum simulations against analytical classical solutions across 150 test runs and implemented experiments on IBM Quantum and IonQ hardware.

Research Intern for LIP (CMS Collaboration from CERN) *Jun 2025 - Sep 2025*

- Developed C++/ROOT analysis tools for CMS Run 3 collision data, using RooFit maximum-likelihood fits, background subtraction and efficiency corrections to study B-meson production.
- Evaluated Boosted Decision Tree (BDT) discrimination using ROC curves and significance-based cut optimization, alongside statistical and systematic uncertainty analysis.
- Presented research findings twice to the CMS Collaboration and co-authored an internal publication and analysis note; additionally contributed to X(3872) studies.

Open-Source Contributions

Shortcut Sensitivity in Agent-Safety Evaluations *Oct 2026*

- Contributed a merged post-publication extension to Shortcut Sensitivity in Agent-Safety Evaluations, analyzing AgentDojo weighting with grouped cross-validation, domain holdouts and bootstrapping.

Projects

1st Cycle Integrated Project in Engineering Physics, ML4Plasmas *Feb 2026 - Jun 2026*

- Developed and trained PyTorch neural networks to estimate oxygen-plasma reaction-rate coefficients from LoKI simulations, investigating inverse modeling and ensemble methods in a project graded 19/20, supervised by Professor Vasco Guerra and Professor Rodrigo Ventura.
- Evaluated species selection, neural network ensembles and robustness to noisy inputs, achieving $\sim 10^{-6}$ scaled test MSE on noise-free data and identifying noise sensitivity as a key limitation.

Traffic Lights Q-Agent with Reinforcement Learning *Aug 2025 - Sep 2025*

- Developed a tabular Q-learning agent in Python using ε-greedy exploration and temporal-difference updates, supporting self-play and configurable multi-opponent training curricula.
- Implemented evaluation against random, tactical and minimax-based opponents with alpha-beta pruning, alongside interruption-safe training checkpoints and an interactive Tkinter interface.

Awards & Honors

LEFT/MEFT Student Innovation Prize *Sep 2025*

- Awarded the LEFT/MEFT Student Innovation Prize at the IST-Physics Frontier Symposium, with the Noisr team, along with a monetary prize in recognition of our project on Physical Neural Networks.

Academic Merit Diploma (2024/25)